



SUPPLY CHAIN RESILIENCE TRACK

Supply Chain Disruption Assistant & Fleet Utilization Optimizer

Turning chaotic disruptions into coordinated, cost-saving decisions — in real time.

TEAM-FUTURE

The Problem

Disruptions cascade faster than teams can track them manually



Cascading disruptions

Weather events, port strikes, and geopolitical crises ripple across hundreds of active shipments at once — far beyond what dispatchers can track by hand.



Idle & overloaded fleets

While some routes sit gridlocked, trucks, containers, and vessels elsewhere sit idle — capacity goes to waste in both directions.



Cold chain blind spots

Vaccines and perishables are especially fragile. A single temperature excursion on any leg of the journey can spoil cargo worth \$500K or more.



DISCOVERED TOO LATE

\$500K+

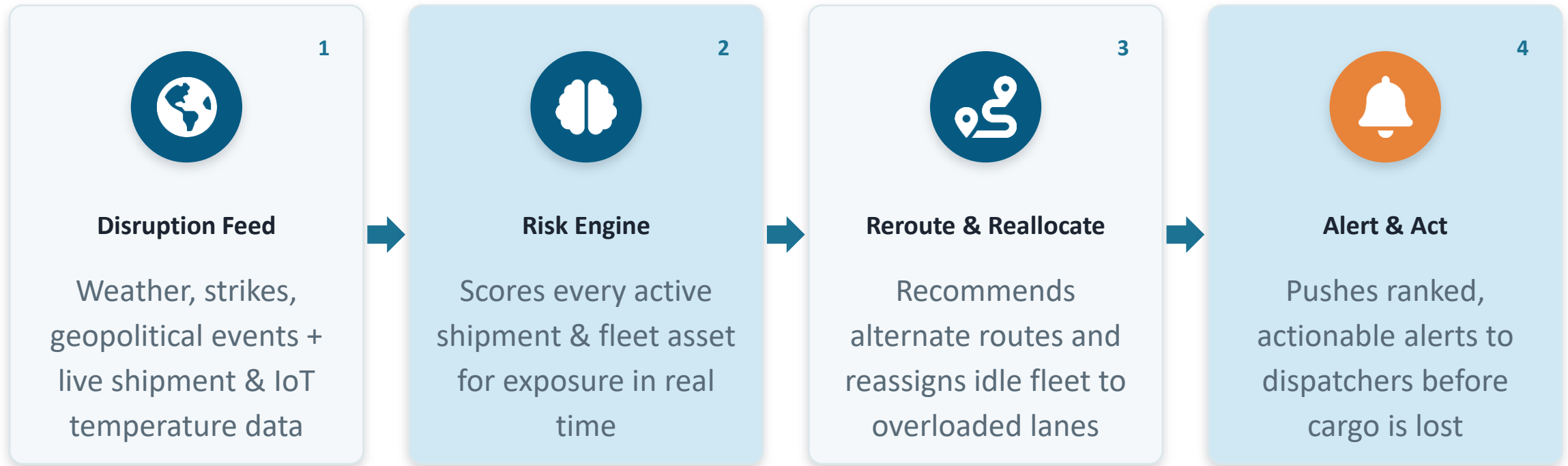
in cold-chain cargo value at risk per breach — and today, breaches are typically found only at delivery.

WHO FEELS IT

Logistics operators, fleet dispatchers, and pharma / cold-chain shippers who need to act within minutes, not after delivery.

Our Solution

One pipeline: sense disruption → score risk → recommend action → alert the right people



Result: disruptions are caught upstream, fleet capacity is rebalanced automatically, and cold-chain cargo is protected before spoilage occurs.

Shipment Manager

**Supply Chain Assistant**
Fleet Utilisation Optimizer

CONTROL TOWER

 Dashboard

 Alerts & Notifications

 Disruption Mgmt

1

 **Shipment Manager**

 Route Optimisation

 Carrier Selection

 Fleet Utilisation

 Cold-Chain Monitor

7

 AI Investigation

 Action Centre

12

 Activity Log

11:50:06 pm Updated 11:49:58 pm

 Sim: initialising...

 Simulated Live Data

 Stop

 Pause

 Resume

 57



 Reset

 **Shipment Manager**

Monitor and manage all active shipments with live disruption impact data

 Search ID, origin, destination, cargo...

All Priorities ▾

All Risk Levels ▾

All Statuses ▾

All Cargo Types ▾

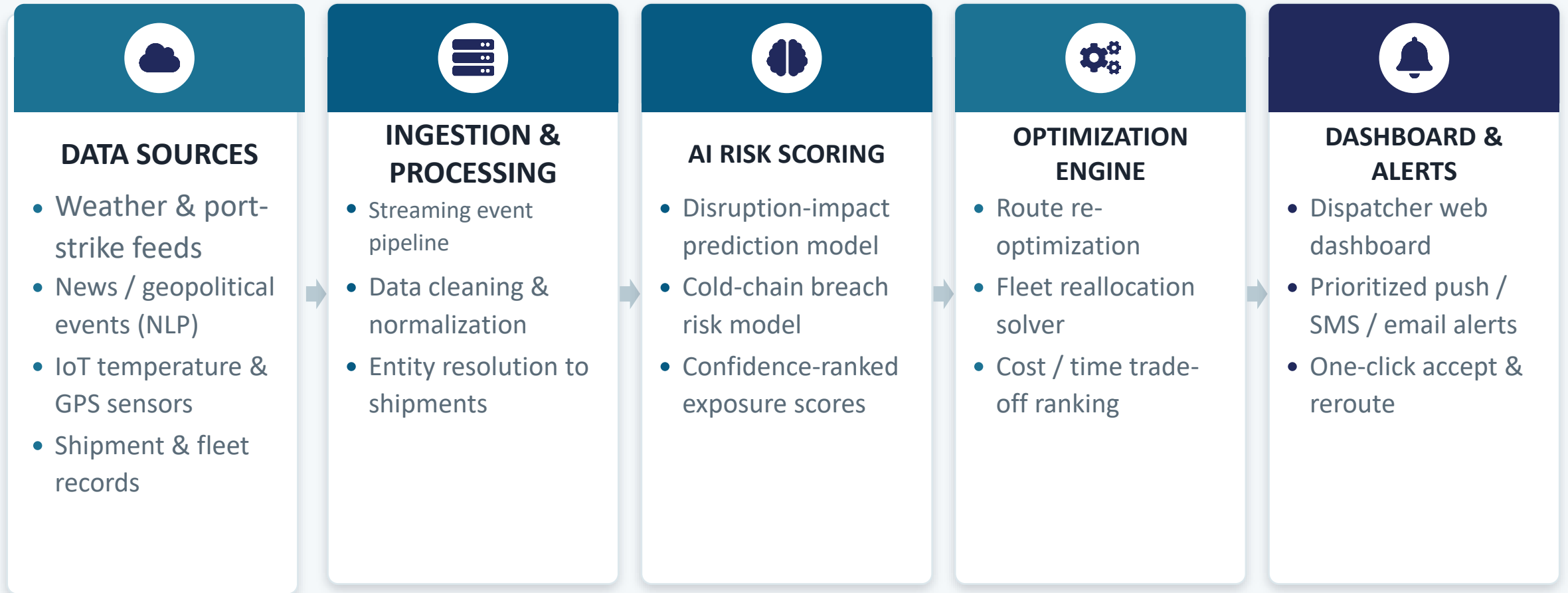
Sort: ID ▾

ID	ORIGIN → DESTINATION	CARGO	CARRIER	ETA	DELAY	PRIORITY	RISK	TEMP
SHP-001	Houston, TX → Atlanta, GA	Pharmaceuticals ❄️	OceanFreight Co	2025-07-16 14	—	CRITICAL	LOW	2-8°C
SHP-002	Los Angeles, CA → Chicago, IL	Electronics	PacificShip Lines	2025-07-17 09	+12h	HIGH	HIGH	15-25°C
SHP-003	Laredo, TX → Dallas, TX	Automotive Parts	BorderXpress	2025-07-15 11	—	HIGH	LOW	Ambient
SHP-004	Denver, CO → New York, NY	Retail Goods	MidWest Haulers	2025-07-18 16	—	MEDIUM	LOW	Ambient
SHP-005	Long Beach, CA → Seattle, WA	Consumer Electronics	WestCoast Freight	2025-07-16 08	—	HIGH	LOW	10-30°C
SHP-006	Phoenix, AZ → Albuquerque, NM	Fresh Produce ❄️	SunState Carriers	2025-07-15 07	+11h	CRITICAL	HIGH	4-10°C



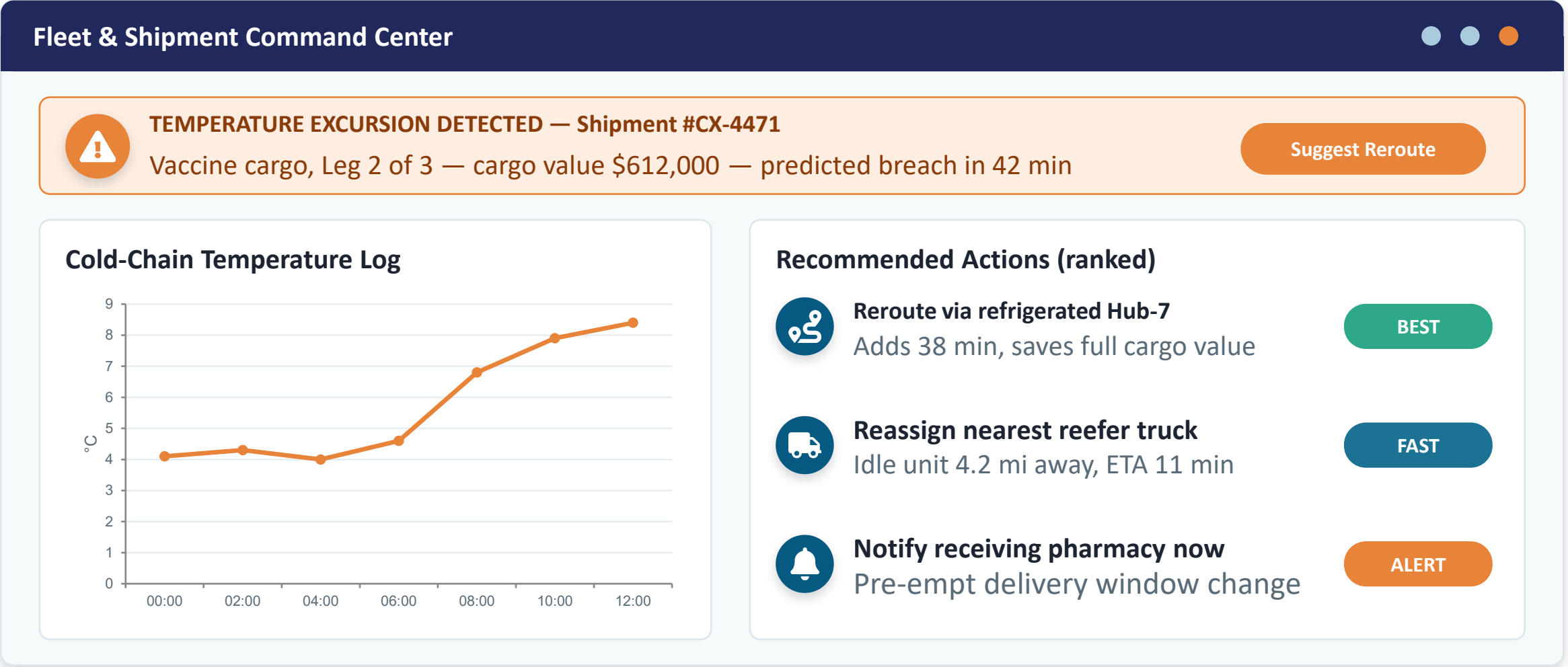
Architecture

A layered pipeline from raw signals to dispatcher action



Demo: Cold-Chain Breach Prevention

Our strongest feature — catching a temperature excursion before it becomes a \$500K loss



IBM Technologies

[Replace with the exact IBM products / services your build actually used]



watsonx.ai

Trains and serves the disruption-impact and cold-chain breach risk-scoring models.



Watson Discovery / NLP

Extracts disruption events (strikes, weather, unrest) from live news and reports.



IBM Db2 / Cloudant

Stores shipment, fleet, and sensor telemetry with low-latency reads for scoring.



IBM Cloud Functions

Serverless orchestration connecting ingestion, scoring, and alerting stages.



IBM Maximo

Tracks fleet asset status and utilization feeding the reallocation engine.



IBM Cloud Security

Identity, access control, and encryption across the data and API layers.

AI Investigation Brief

CONTROL TOWER

-  Dashboard
-  Alerts & Notifications
-  Disruption Mgmt

1
-  Shipment Manager
-  Route Optimisation
-  Carrier Selection
-  Fleet Utilisation
-  Cold-Chain Monitor

8
-  AI Investigation
-  Action Centre

12
-  Activity Log

 AI Investigation Brief

Rule-based intelligent summary of the current supply chain situation



AI Investigation Brief

Intelligent situation summary · IBM BOB AI Agent Demo

AI AGENT DEMO

PRIORITY: CRITICAL

 This is a simulated AI-generated recommendation using rule-based logic and synthetic data. Not a real AI model output.

1. SITUATION SUMMARY

A **Medium Infrastructure** disruption is currently active at **I-80 Nebraska Corridor**. 4 of 12 tracked shipments are affected, generating an estimated 33 hours of cumulative delay. Estimated \$450K delay cost. Auto parts and retail goods affected.

2. MOST AFFECTED SHIPMENTS

- **SHP-004** – Denver, CO → New York, NY | Retail Goods | **+14h delay** | Medium priority
- **SHP-006** – Phoenix, AZ → Albuquerque, NM | Fresh Produce | **+9h delay** | Critical priority
- **SHP-009** – Omaha, NE → Cleveland, OH | Auto Parts | **+7h delay** | Medium priority

3. ROOT CAUSE ANALYSIS

Primary cause: **Highway Closure** (Infrastructure) at I-80 Nebraska Corridor. This is a Medium-severity event expected to last 24–36 hours. Bridge structural damage on I-80 near Grand Island, NE. Closed indefinitely pending inspection.

4. DELAY PREDICTION



Results & Impact

[Replace illustrative figures below with your actual pilot / test results]



80%

Faster disruption detection
vs. manual monitoring
baseline



92%

Breach prediction accuracy
on held-out cold-chain test
set



35%

Less idle fleet time
through automated
reallocation



\$2.3M

Cargo value protected
in simulated pilot scenarios

WHAT SUCCESS LOOKS LIKE

- No cold-chain shipment reaches its destination without an early temperature-risk warning.
- Idle fleet capacity is matched to overloaded lanes within minutes of a disruption event.
- Dispatchers get one ranked action, not a flood of raw alerts, cutting decision time to seconds.

The Team

[Replace placeholder names, roles, and contributions]



Ansh Patel

ML & Risk Modeling

Built the disruption-impact and cold-chain breach prediction models using watsonx.ai.



Jeet patel

Frontend & Dashboard

Designed and built the dispatcher command center and alert workflow UI.



Kshitij Pandya

Data Engineering

Built the ingestion pipeline connecting weather, news, IoT, and shipment data.



Ansh Patel

Cloud & DevOps

Set up IBM Cloud infrastructure, serverless orchestration, and deployment.

Supply Chain Assistant
Fleet Utilisation Optimizer

11:42:28 pm Updated 11:42:28 pm

Sim: initialising...

Simulated Live Data

Stop

Pause

Resume

57

Reset

CONTROL TOWER

Dashboard

Alerts & Notifications

Disruption Mgmt 1

Shipment Manager

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Activity Log

TOTAL SHIPMENTS

12

→ Active consignments

ACTIVE DISRUPTIONS

1

↑ Select to simulate

AFFECTED SHIPMENTS

3

↑ Disruption impact

HIGH PRIORITY

9

→ Critical & High

FLEET UTILISATION

34%

→ Avg across all assets

IDLE ASSETS

3

→ Ready for redeployment

COLD-CHAIN ALERTS

7

↑ Unacknowledged sensors

TOTAL DELAY HOURS

48h

↑ Cumulative impact

FINANCIAL RISK

\$153K

↑ 12% of cargo value

PENDING ACTIONS

10

→ Require attention

Shipment Status Distribution

Current status across all 12 shipments



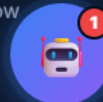
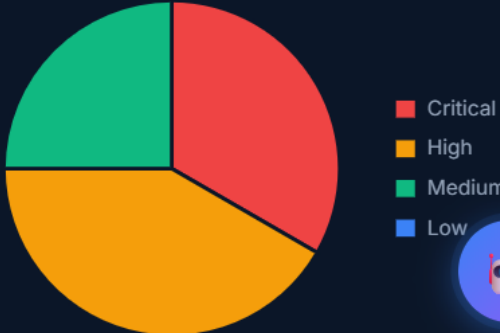
Fleet Utilisation

Load percentage per vehicle



Priority Distribution

Shipments by priority level



Real-time operational breaches across temperature, delays, disruptions, missed ETAs, idle fleet, capacity, and high-risk cargo.

Archived resolution

Search alerts by shipment, asset, or title

open

☒ **Resolved**



Alert and Notification Control Tower

**Supply Chain Assistant**
Fleet Utilisation Optimizer

CONTROL TOWER

Dashboard

Alerts & Notifications

Disruption Mgmt

Shipment Manager

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Activity Log

11:44:33 pm Updated 11:44:28 pm

Sim: initialising...

Simulated Live Data

**Alert & Notification Control Tower**
Real-time operational breaches across temperature, delays, disruptions, missed ETAs, idle fleet, ca

Re-evaluate Engine

TOTAL SYSTEM ALERTS
58
Across 7 breach types

ACTIVE CRITICAL BREACHES
22
Immediate action needed

ACTIVE WARNINGS
36
SLA & capacity alerts

Severity: All Severities Status: All Statuses Category: All 7 Categories

CRITICAL

High-Risk Cargo Vulnerability: SHP-001

Sensitive consignment SHP-001 (Pharmaceuticals) is currently at risk with high loss exposure.

ID ALT-058 SHP-001 2026-09-14 17:16 UTC

Acknowledge Resolve

WARNING

Missed ETA SLA

Shipment SHP-001 customer delivery

ID ALT-057 SHP-001

Acknowledge Resolve

Affected shipments

Delayed

Disruption

Idle fleet

Cold-chain

Action plan

Alt route

Risk

Go to Action Centre to complete or snooze tasks.

11:44 pm

Which shipments are affected?

11:44 pm

Ask me any operational query using live database data:

- "Which shipments are at risk?"
- "Which trucks are idle?"
- "Which cold-chain shipment has a temperature breach?"
- "How can we reduce today's delays?"

11:44 pm

Ask about shipments, routes, fleet...

Acknowledge Resolve

Cold-Chain Monitor

IoT sensor readings for temperature-sensitive shipments

- All Sensors
- Critical
- Warning
- Normal

SHP-001 – SN-A1

Pharmaceuticals · Houston, TX → Atlanta, GA

NORMAL

7°C

Allowed: 2°C to 8°C

Humidity

68%

Battery

80%

Door

Closed

GPS

32.1°N, 95.8°W

Timestamp

2026-09-14 18:15

SHP-006 – SN-B2

Fresh Produce · Phoenix, AZ → Albuquerque, NM

CRITICAL

14°C

Allowed: 4°C to 10°C

Humidity

74%

Battery

48%

Door

Closed

GPS

33.4°N, 112.0°W

Timestamp

2026-09-14 18:15

✓ Acknowledge

↑ Escalate

🔍 Inspect

SHP-008 – SN-C3

Food & Beverage · Oakland, CA → Las Vegas, NV

CRITICAL

5.2°C

Allowed: 0°C to 4°C

Humidity

90%

Battery

37%

Door

Closed

GPS

37.8°N, 122.3°W

Timestamp

2026-09-14 18:15

✓ Acknowledge

↑ Escalate

🔍 Inspect

SHP-010 – SN-D4

Chemicals · Houston, TX → Miami, FL

CRITICAL

SHP-011 – SN-E5

Medical Supplies · San Diego, CA →

NORMAL

SHP-012 – SN-F6

Perishable Goods · McAllen, TX → Kansas

CRITICAL



CONTROL TOWER

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 **Fleet Utilisation**

 **Cold-Chain Monitor** 7

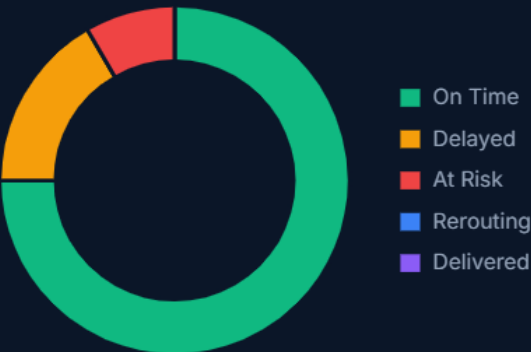
 **AI Investigation**

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 **Activity Log**

Shipment Status Distribution

Current status across all 12 shipments



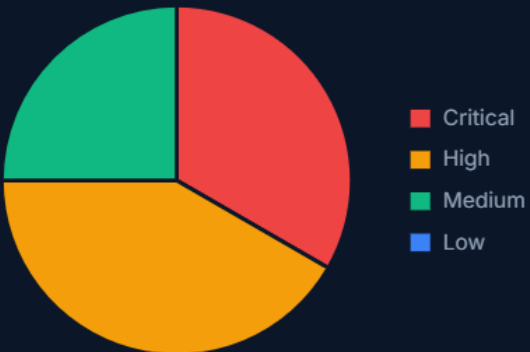
Fleet Utilisation

Load percentage per vehicle



Priority Distribution

Shipments by priority level



Cold-Chain Temperature Readings

Sensor readings vs. allowed range

